15. Symmetry-indicator algebra and the limits of band diagnosis

Pure Thought research portfolio

Revised 17 September 2026

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Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

Symmetry can act like a fingerprint for quantum bands: it provides a compact set of labels that can sometimes reveal nontrivial topology. But different objects can share a fingerprint. This problem asks exactly what can be inferred from symmetry labels, how to compute that inference reliably, and when the correct answer is that more information about the quantum states is needed.

Key terms

- **Crystal symmetry** — A translation, rotation, reflection or related operation preserving the lattice structure and the stated physical setting.
- **Representation** — Matrices describing how quantum states transform under a symmetry.
- **Irreducible representation (irrep)** — A basic representation that cannot be decomposed into smaller independent representations of the same group.
- **Compatibility relations** — Rules requiring symmetry labels at different momenta to fit into continuous bands.
- **Atomic band representation** — Band data generated from suitably localized, symmetry-compatible orbitals.
- **Wannier function** — A localized real-space representation of band states; the existence of an appropriate symmetry-respecting basis is central to atomicity.
- **Symmetry indicator** — An algebraic label detecting certain obstructions to atomic band data, but not every possible topological distinction.
- **Smith normal form** — An exact integer-matrix decomposition used to calculate the quotient of symmetry-data lattices.
- **Fragile topology** — An obstruction that can disappear when suitable trivial bands are added, under the chosen definition.

Why this matters

Symmetry labels are often much simpler to work with than the full wavefunctions across momentum space. Knowing what those labels do and do not prove can make topological diagnosis more efficient and prevent false declarations that a system is ordinary. The distinction is essential for turning a classification table into a trustworthy scientific tool.

What progress would mean

A useful result would be an exact indicator calculation for one symmetry setting, together with examples showing its reach and blind spots. Stronger conclusions about a particular Hamiltonian would use additional information, such as Berry phases or a Wannier construction.

Core question

For one fixed symmetry setting, can compatibility relations and atomic band representations be generated exactly, and what topology can their quotient actually detect?

Scope and assumptions

Choose one small symmetry setting, initially a spinless two-dimensional inversion-symmetric lattice without time reversal. Specify the space-group generators, orbital/site representations, and high-symmetry momenta. Derive all input representations from group operations, with external tables used only as checks.

The result is an algebraic classification of symmetry data. A zero indicator is not a certificate of a trivial occupied bundle.

Mathematical target

Let $B \subset \hat{Z}r$ be the lattice of integer irrep vectors satisfying compatibility constraints, and let $A \subset B$ be the integer lattice generated by atomic band-representation vectors. Compute $X = B/A$ by Smith normal form, including basis-change matrices.

For a physical nonnegative irrep vector b, distinguish integer-lattice membership from nonnegative decomposition b=A_mat n, $n \geq 0$. Nonnegative decomposition establishes compatibility with atomic symmetry data, not by itself a Wannier construction for a supplied Hamiltonian. Conversely, an actual symmetry-preserving localized Wannier basis is constructive atomic evidence.

Required outputs and proof obligations

Export group conventions, irreps, compatibility matrix, atomic generators, Smith decomposition, indicator mapping, and witnesses for any integer/nonnegative decomposition. Report "indicator trivial; topology unresolved" when symmetry data are insufficient. Use Berry/Wilson-loop information when making a stronger claim about a specific H.

Validation and rejection controls

Verify all representation identities and dimension counts. Reconstruct the Smith identity U A_mat V=D in the chosen B basis, with U,V unimodular. Test known atomic data, incompatible data, and models sharing symmetry labels but differing in an additional invariant. Keep stable and fragile notions distinct.

Milestones and research extension

Baseline: exact quotient and decompositions for one setting. Strong: connect the diagnosis to explicit Hamiltonians and identify its kernel. Research extension: another symmetry group or a rigorous fragile obstruction with a definition of allowed added bands.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- Po, Vishwanath and Watanabe, Symmetry-based Indicators of Band Topology — Quotient of compatible symmetry data by atomic data; symmetry-based diagnosis.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

$$A \subset B \subset \mathbb{Z}^r, \qquad X = B/A, \qquad UA_{\mathrm{mat}}V = D, \quad U, V \text{ unimodular.}$$

$$b = A_{\mathrm{mat}}n, \quad n \in \mathbb{Z}^s \qquad \text{versus} \qquad b = A_{\mathrm{mat}}n, \quad n \in \mathbb{Z}^s_{\geq 0}.$$

Use a declared basis for the compatibility lattice B in the Smith decomposition. Integer membership and nonnegative atomic symmetry-data decomposition are distinct tests. Neither a zero indicator nor matching atomic labels alone constructs localized Wannier functions for a given Hamiltonian.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Blocking — "complete classification" (14–22, 36–44).** Symmetry indicators generally do not distinguish every topological bundle. Indicator-trivial phases can still be topological.
2. **Blocking — cone versus group (79–105).** The text interchanges nonnegative EBR sums, integer differences, and a group quotient. Fragile topology and stable indicator obstructions require these to be kept separate.
3. **Major — zero indicator implies positive decomposition (103–105).** Integer-lattice membership does not guarantee a nonnegative solution. Even matching an atomic irrep vector is not a construction of localized Wannier functions for a particular Hamiltonian.

4. **Major — crystallography input.** Not every site representation induction is automatically elementary; site symmetry, maximality and spin/time-reversal conventions must be stated. The revised algebra produces auditable limited conclusions instead of an all-phase label.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **For one fixed symmetry setting, can compatibility relations and atomic band representations be generated exactly, and what topology can their quotient actually detect?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: exact quotient and decompositions for one setting. Strong: connect the diagnosis to explicit Hamiltonians and identify its kernel. Research extension: another symmetry group or a rigorous fragile obstruction with a definition of allowed added bands.

Sources supporting the corrections

- Po, Vishwanath and Watanabe, Symmetry-based Indicators of Band Topology — Quotient of compatible symmetry data by atomic data; symmetry-based diagnosis.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.

- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.